ENGO 419 GEOMATICS NETWORKS

LAB #6: NETWORK RELIABILITY MEASURES

OBJECTIVES: To examine the output of a Least-Squares Network Adjustment and determine internal and external reliability values

DATE: Wednesday, 21 October 2026

DATE DUE: Friday, 23 October 2026 (6:00 PM)

This lab may be completed and submitted in groups of one, two, three or four. Only one group member should submit the brief report.

Introduction:

Last lab you visualized the results of a network adjustment using C_x to draw error ellipses. This lab you will compute reliability measures for each observation.

Task 1: Setup

a) Again load the final $\hat{x}$, and the $C_{\hat{x}}$ from Lab 4 and compute the covariance matrix of the residuals C_r (or C_v)

b) Compute the redundancy number of each observation.

c) Calculate the minimum detectable blunder (MDB) for each observation using $\alpha = 0.05$ and $\beta = 0.20$.

d) Calculate the corresponding external reliability vector $\nabla\hat{x}_i$ for each MDB, assuming that only one minimum detectable blunder occurs at a time. Recall that:

$$\nabla\hat{x}_i = (A^T C_l^{-1} A)^{-1} A^T C_l^{-1} \nabla l_i$$

where ∇l_i is a vector containing one MDB in the i-th row with all other elements zero. Note that each MDB generates a full vector $\nabla\hat{x}$ representing the impact of that that one MDB on the whole solution.

For each $\nabla\hat{x}$ identify the three largest components, ie of all the parameters, the three that would be most impacted by that particular MDB.

e) Task d) generated a lot of results. A simpler way to generalize this result is to take a weighted sum-of-squares of the elements of $\nabla\hat{x}_i$. (We did not discuss this in class). This can be done by calculating $\nabla\hat{x}^T C_{\hat{x}}^{-1} \nabla\hat{x}$ and in some references is called the global external reliability measure.

f) Present your results in a table in your report that includes the observation, redundancy number, MDB, global external reliability measure, and the three largest elements of each $\nabla\hat{x}_i$ (name which parameter and give the value). Comment on the results. Which parts of the network have better reliability, which parts are worse.

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MATERIAL TO BE SUBMITTED

1. A report following the simple template posted on D2L that includes:
2. The results obtained from each of the tasks including answers to questions asked in the lab, a table of results for each observation.
3. A copy of any new or modified source code used to generate the results with sensible variable names and brief comments. You do not have to resubmit your network adjustment from Labs 3 or 4, you can start with simply loading those results. As always, you may use software developed by you for ENGO 363, but make sure to say so in the comments.

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